Platform for Habitability Assessment and Exploration

Author: Meet Bharatbhai Desai
Affiliation: B.Tech Computer Engineering, SAL Institute of Technology & Engineering Research, Ahmedabad, India
Email: amrmeetdesai02432@gmail.com

Abstract

The search for intelligent environmental assessment systems has become increasingly important for space exploration, disaster management, and remote environmental monitoring. Traditional monitoring systems primarily focus on collecting sensor data but lack the capability to perform autonomous environmental interpretation. This paper presents MirageMind AI, an autonomous environmental intelligence platform designed to analyze environmental conditions and estimate habitability using sensor fusion and intelligent decision-making techniques. The proposed system integrates temperature, humidity, gas concentration, light intensity, and obstacle detection sensors with an ESP32-based processing unit. The collected environmental data is analyzed through a habitability assessment engine that generates a Life Possibility Score and classifies environments as hostile, potentially habitable, or habitable. A real-time radar visualization dashboard enables environmental monitoring and decision support. The project demonstrates the feasibility of combining embedded systems, robotics, and artificial intelligence for future intelligent exploration missions.

Keywords: Artificial Intelligence, Robotics, Sensor Fusion, ESP32, Habitability Assessment, Space Exploration, Environmental Intelligence

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1. Introduction

Exploration of remote and unknown environments presents significant challenges due to the lack of real-time environmental understanding. Whether in planetary exploration, hazardous industrial sites, disaster zones, or inaccessible terrains, environmental intelligence plays a critical role in decision-making.

Current sensing systems often provide raw sensor data without performing autonomous interpretation. Human operators are required to analyze environmental conditions manually, which increases response time and limits operational efficiency.

MirageMind AI aims to address this challenge by integrating multiple environmental sensors with intelligent data processing algorithms capable of estimating habitability and environmental risks. The proposed system uses sensor fusion techniques to combine environmental parameters and generate a comprehensive environmental assessment.

The long-term vision of MirageMind AI is to support autonomous robotic exploration systems capable of assisting future planetary exploration missions, environmental monitoring operations, and disaster-response applications.

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2. Problem Statement

Future exploration missions require systems capable of understanding environmental conditions autonomously. Traditional sensor systems provide individual measurements but fail to evaluate whether an environment can potentially support life or safe exploration activities.

Major challenges include:

- Lack of intelligent environmental interpretation.
- Limited integration of multiple environmental sensors.
- Absence of autonomous habitability assessment.
- Difficulty in identifying potentially hazardous environments.
- Need for real-time decision support systems.

MirageMind AI addresses these challenges through environmental intelligence and sensor fusion techniques.

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3. System Architecture

The proposed MirageMind AI system consists of the following major components:

3.1 Sensor Layer

Environmental parameters are collected using:

- HC-SR04 Ultrasonic Sensor
- DHT22 Temperature and Humidity Sensor
- MQ135 Air Quality Sensor
- LDR Light Sensor

These sensors provide environmental information required for habitability analysis.

3.2 Data Acquisition Layer

An ESP32 microcontroller acts as the primary data acquisition and processing unit. Sensor readings are collected and transmitted to the visualization system.

3.3 Sensor Fusion Layer

Environmental parameters are combined using sensor fusion techniques to create a unified representation of environmental conditions.

3.4 Habitability Assessment Engine

The habitability assessment engine evaluates environmental parameters and generates a Life Possibility Score.

3.5 Decision Engine

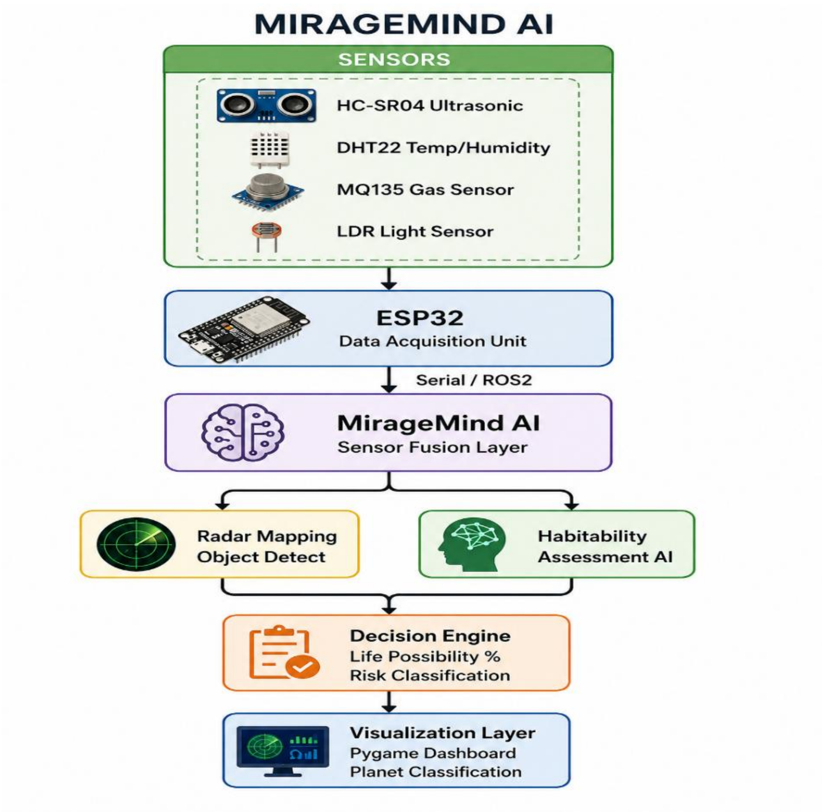
Based on the calculated score, the environment is classified as:

- Hostile Environment
- Potentially Habitable Environment
- Habitable Environment

3.6 Visualization Layer

A Python-based dashboard provides:

- Radar Visualization
- Sensor Monitoring
- Environmental Status Display
- Life Possibility Percentage



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4. Methodology

The proposed methodology consists of four major stages.

4.1 Environmental Data Collection

Sensor data is continuously collected from:

- Temperature Sensor
- Humidity Sensor
- Air Quality Sensor
- Light Sensor
- Distance Sensor

4.2 Sensor Fusion

Multiple environmental parameters are combined to create a comprehensive environmental profile.

4.3 Life Possibility Assessment

A rule-based algorithm evaluates environmental conditions using predefined thresholds.

The algorithm considers:

- Temperature suitability
- Humidity suitability
- Air quality conditions
- Light availability

Each parameter contributes to the final Life Possibility Score.

4.4 Environmental Classification

Based on the calculated score:

- 0–40% → Hostile
- 41–75% → Potentially Habitable
- 76–100% → Habitable

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5. Implementation

The system is implemented using both hardware and software components.

Hardware

- ESP32 Development Board
- HC-SR04 Ultrasonic Sensor
- DHT22 Sensor
- MQ135 Sensor
- LDR Sensor
- SG90 Servo Motor

Software

- Arduino IDE
- Python
- Pygame
- ESP32Servo Library
- DHT Library

The ESP32 collects environmental information and transmits sensor data through serial communication. A Python-based dashboard receives the information and visualizes environmental conditions in real time.

6. Experimental Results

Several environmental scenarios were simulated to evaluate the performance of MirageMind AI.

Parameter	**Sample Value**
Temperature	28°C
Humidity	58%
Gas Level	450
Light Intensity	1800
Distance	50 cm
Life Possibility Score	85%
Classification	Habitable

The system successfully generated environmental classifications and displayed results through the dashboard.

The experimental results indicate that the proposed architecture can effectively evaluate environmental conditions using low-cost embedded hardware.

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7. Applications

MirageMind AI has potential applications in:

- Planetary Exploration
- Mars Rover Systems
- Lunar Exploration Missions
- Environmental Monitoring
- Disaster Management
- Industrial Safety Monitoring
- Autonomous Robotics
- Smart Agriculture
- Remote Terrain Exploration

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8. Future Scope

Future development of MirageMind AI may include:

- ROS2 Integration
- OpenCV-Based Terrain Analysis
- Autonomous Rover Deployment
- Machine Learning-Based Habitability Models
- Cloud-Based Data Analytics
- Real-Time Satellite Integration
- Multi-Planet Simulation Environment
- Autonomous Exploration Decision Systems

These improvements will significantly enhance system intelligence and exploration capabilities.

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9. Conclusion

This paper presented MirageMind AI, an autonomous environmental intelligence platform designed for habitability assessment and environmental monitoring. The system integrates environmental sensing, sensor fusion, and intelligent decision-making techniques to evaluate environmental conditions and estimate the possibility of life-supporting environments. Experimental results demonstrate the feasibility of implementing environmental intelligence using low-cost embedded hardware and software tools. MirageMind AI establishes a foundation for future intelligent exploration systems and autonomous robotic missions.

References

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